Isselmou Oumar

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EXPERIENCE

Société Générale – Quantitative Analyst, Credit Risk Modelling — Paris | Jan 2025 – Present

- Built an **LLM-based news-analytics pipeline** on GDELT: retrieved issuer news over the **180 days preceding each default** and scored climate/ESG sentiment to assess whether ESG factors materially contributed to financial distress; **now used by credit risk teams**
- Parallelised the Monte-Carlo credit portfolio simulation engine (R/C++ via RcppParallel) for **climate stress testing** across 3 NGFS scenarios (Below 2°C, Net Zero 2050, Hot House World) and 3 regions (Europe, Americas, Asia), cutting full-book runtime by **40%**
- Led an exploratory R&D track on embedding **climate transition risk into PD models** for the **€142B corporate book**, formulating default dynamics as a stochastic-control / dynamic-programming problem driven by macro factors and transition-risk signals

Amundi Technology – Quantitative Research Intern — Paris | Apr 2024 – Oct 2024

- Co-authored research on **carbon-price shock transmission to credit markets**; calibrated structural and ML models to CDS-implied spreads to identify mispricing across issuer credit curves
- Extracted **market-implied transition probabilities** by embedding probabilistic scenario trees in a DCF framework, pricing climate-policy tail risk
- Built sector-level transition-risk heat-maps (Python) used to re-weight low-carbon tilt portfolios

SELECTED PROJECTS

- **Day-ahead power price forecasting & battery arbitrage (Python)** – hourly German (DE-LU) prices from archived weather forecasts with gate-closure-consistent features; gradient boosting cuts MAE by **45%** vs. the weekly naive benchmark over 667 out-of-sample days (Diebold–Mariano $p < 0.001$); forecast-driven battery strategy adds **+9.5% P&L** over a naive forecast, net of degradation costs (block-bootstrap 95% CI excludes zero) [code]

Ensimag projects, co-authored:

- **American option pricing – Longstaff–Schwartz (C++)** – LSM on a multidimensional Black–Scholes model with Hermite-polynomial regression; direct and re-simulation (low-biased) estimators for basket, best-of and geometric put options [code]
- **Monte-Carlo pricing & delta hedging of exotics (C++)** – prices and finite-difference deltas with estimator standard deviations for Asian, basket and performance options; discrete delta-hedging backtest reporting hedging P&L [code]
- **Multi-cash-flow derivatives hedging (C++ / C#, gRPC)** – C++ Monte-Carlo pricing server and C# hedging engine with rebalancing oracle, on real calendar dates [code]
- **Replicating-portfolio backtester (C#/.NET)** – self-financing hedge of a derivative on historical market data with configurable rebalancing schedules [code]
- **Swaption demand forecasting – industry project with Crédit Agricole CIB** – Random Forest / SVM models on U.S. insurer data (Python/SQL)

EDUCATION

Grenoble INP – Ensimag – Diplôme d'Ingénieur (MSc), Financial Engineering, *with Honors* — 2022 – 2024
Grenoble INP – IAE – MSc Quantitative Finance (double degree), *with Honors* — 2023 – 2024
Université Grenoble Alpes – BSc Applied Mathematics & Computer Science, *with High Honors* — 2019 – 2022

Relevant coursework: stochastic calculus, numerical methods for finance, statistics, machine learning

SKILLS

Programming: Python (NumPy, pandas, scikit-learn), C++, R (Rcpp), SQL, C#/.NET, Bash; gRPC, CMake, Git, Docker
Quantitative: derivatives pricing & calibration, Monte-Carlo simulation, credit spread & default modelling, statistical & ML modelling, time-series analysis, delta hedging & replication back-testing
Languages: Arabic (native), French (fluent), English (fluent), Spanish (intermediate)

ADDITIONAL

Youth football coach (ages 8–14) · Member, Mauritanian Engineering Committee